

WIRELESS NETWORKING

Wi-Fi Standards and Wireless Networking

1. Wi-Fi Standards (Simple)

Standard	Key Difference
802.11n (Wi-Fi 4)	Up to ~600 Mbps, works on 2.4 GHz and 5 GHz
802.11ac (Wi-Fi 5)	Faster (~1–3 Gbps), mainly uses 5 GHz band
802.11ax (Wi-Fi 6)	Even faster + better efficiency, works on 2.4 GHz + 5 GHz

2. Simple Home Wi-Fi Diagram

Smartphone

| (Wi-Fi - Wireless)



Wi-Fi Router — LAN Cable — ISP Modem — Internet

(Home Network)

Connection Types

- **Wireless Part:** Smartphone → Router
- **Wired Part:** Router → Modem → Internet

3. Example Wi-Fi Scan (Sample Format)

(You can check this on your phone/laptop Wi-Fi settings)

SSID	Signal Strength	Security
Home WIFI	Strong	WPA2
JioFiber_5G	Medium	WPA3
Neighbor Net	Weak	WPA2

Which to Choose?

- **Best Choice:** Home WIFI
- **Reason:** Strong signal + secure (WPA2/WPA3 is safe; open or weak networks are not preferred)

4. SSID Hiding (Simple Explanation)

SSID hiding means the Wi-Fi name is not shown in the available networks list. You must manually enter the network name to connect.

Advantage

- Slightly improves privacy (random users can't easily see your Wi-Fi name)

Disadvantage

- Not very secure (hackers can still detect it)
- Harder for users to connect (you must type SSID manually)